

No influence of melatonin on cerebral blood flow in humans.

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Melatonin has been attributed a role in a number of physiological processes. Changes in distal skin temperature and blood pressure after intake of melatonin suggest that melatonin induces peripheral vasodilation. The effect on the cerebral blood flow is still unknown. We examined the effect of a single pulse of melatonin on cerebral and peripheral blood flow, using the latter as a positive control. Ten male healthy volunteers (mean age, 22 +/- 3.2 yr) participated in a double-blind, randomized, placebo-controlled, cross-over study. On one occasion 10 microg melatonin were infused i.v., and on the other occasion saline was infused as the matching placebo. Cerebral blood flow was measured using phase contrast magnetic resonance imaging. Peripheral blood flow was determined from changes in the distal to proximal skin temperature gradient and finger pulse volume. Serum melatonin concentration increased from 12 +/- 5 pg/ml at baseline to 487 +/- 377 pg/ml at 5 min and 156 +/- 68 pg/ml at 10 min after melatonin administration. There was no significantly different time course for cerebral blood flow and cerebrovascular resistance. Compared with placebo, melatonin significantly increased peripheral blood flow, as measured by distal to proximal skin temperature gradient and finger pulse volume. These data demonstrate that melatonin does not have an acute regulatory effect on cerebral blood flow in humans.